

Gaps and Overlaps When We Measure

Answer Key

Kindergarten–Grade 1

Quick Answers

- | | | |
|------|------|------|
| 1. 8 | 3. 8 | 5. 4 |
| 2. — | 4. 2 | 6. — |
-

Common wrong answers

3. *If they got 10: reported the number of cubes counted, but the cubes overlapped so two extra cubes were squeezed onto the ribbon*
4. *If they got 18: put the two counts together instead of finding how many more*
5. *If they got 16: added the two numbers instead of finding how many cubes the gaps hid*
-

The numbers behind every picture. The red ribbon is truly **8** cubes long. Ivy laid her cubes end to end and counted 8. Max left gaps and counted 6. Rex overlapped his cubes and counted 10. The yellow straw is truly 10 cubes long; Sam left gaps and counted 6.

1. Count Ivy’s cubes. How long is the ribbon?

Ivy’s cubes touch each other, and none of them sit on top of another — so each cube covers a brand-new piece of ribbon and nothing gets covered twice. That means the number of cubes *is* the length. Touch and count once per cube: 1, 2, 3, 4, 5, 6, 7, 8. The ribbon is 8 cubes long.

Watch for: a child who counts a cube twice, or who skips the last one.

2. Write < or >: Max’s count ___ real length.

Max left gaps. Every gap is a piece of ribbon that no cube covered, so the ribbon ran out of room before Max ran out of counting — he needed *fewer* cubes than the ribbon really holds. His count is 6; the real length is 8. Compare the two numbers: 6 comes before 8 when we count, so 6 is smaller. The symbol always opens toward the bigger number, so it points at the real length: <.

Say it out loud: “Max’s count is less than the real length. Gaps make the number too small.”

3. Find the mistake: Rex counted 10 and said the ribbon is 10 cubes long.

(a) Rex let his cubes *overlap* — each cube climbed on top of the one before it, so part of every cube covered ribbon that was already covered. When cubes overlap, extra cubes squeeze on, and the count grows past the true length: Rex’s 10 is $8 + 2$, the true 8 plus 2 cubes’ worth of double-covering.

(b) The ribbon did not change when Rex measured it. Ivy’s careful lay-down — no gaps, no overlaps — is the one that gives the length, so the ribbon is 8 cubes long.

Full credit for (a) says “the cubes overlapped / sat on top of each other”, not just “he counted wrong”.

4. How many more cubes did Rex count than Ivy?

“How many more” compares two counts, so write both down first: Rex 10, Ivy 8. Line them up one to one — 8 of Rex’s cubes pair with Ivy’s 8, and the leftovers are the extra ones that only fit because of the overlapping. Count on from 8 to 10: 9, 10 — that is two counts. $10 - 8 = 2$, so Rex counted 2 cubes more.

Watch for: adding to get 18. “How many more” is a difference, never a total.

5. Find the mistake: Sam counted 6 cubes on a straw that is really 10.

(a) Sam left *gaps* between his cubes. Those spaces are straw that no cube covered, so the row of cubes reached the end of the straw before Sam had used all the cubes the straw really holds. Gaps make the count too small.

(b) “How many cubes did the gaps hide?” asks for the difference between the true length and Sam’s count: the straw is 10 cubes, Sam counted 6. Count on from 6 to 10: 7, 8, 9, 10 — that is four counts, so $10 - 6 = 4$. The gaps hid 4 cubes.

Watch for: adding to get 16, or answering 6 (Sam’s own count) instead of what is missing.

6. Is Rex right that his bigger number means better measuring? (open response — read the drawing and the reasoning)

The ribbon never changed length. Only the *way the cubes were laid down* changed. Rex’s cubes overlapped, so extra cubes fit on the same ribbon and his count grew to 10. Ivy’s cubes touched with no gaps and no overlaps, so her 8 is the real length. A bigger number can come from sloppy measuring, so it does not mean a better measurement. Therefore Rex is not right.

A full-credit answer draws the ribbon with cubes overlapping and says something like: “Rex’s cubes are on top of each other, so more cubes fit. The ribbon is still 8 cubes long.” *Not full credit:* “Yes, because 10 is bigger than 8” — that compares the counts instead of checking how the cubes were laid.